

Yi Cui

Homepage, GitHub, LinkedIn

EDUCATION

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- **University of North Carolina at Chapel Hill (Ph.D. in Econometrics and Statistics)** Chapel Hill, NC
Research field: Causal Inference, Econometrics (Forecasting), Machine Learning and Deep Learning Sep 2020 - Aug 2025
 - **Fudan University (Bachelor of Arts, Economics)** Shanghai, China
Graduated with Distinction, Outstanding Graduate Student (top 1%) Sep 2016 - Jul 2020
 - **University of California, Los Angeles (Exchange Student, UCEAP program)** Los Angeles, CA
Santander Scholarship (top 1%), Graduate Honor Course: MAE 271A (A) Sep 2017 - Dec 2017

WORK EXPERIENCE

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- **Snap** Palo Alto, CA
Machine Learning Engineer: Content Recommendation Ranking Dec 2025 - Present
 - **Tasks:** Developed and optimized Embedding Based Retrieval (EBR) and Graph-based ML models powering Spotlight and Non-Friend Story content feeds; explored Generative Retrieval and LLM-based recommendation (LLM4Rec) approaches to advance next-generation content retrieval; designed and conducted large-scale A/B experiments to evaluate retrieval model performance and drive data-informed launch decisions; built end-to-end ML pipelines for content retrieval and notification systems encompassing feature engineering, model training, and real-time production serving
 - **Wayfair** Seattle, WA
Economist II: Search, Ads and Product Recommendation Science Jul 2025 - Dec 2025
 - **Tasks:** Developed transformer-based SCUBA model powering "Recommended For You," driving \$16.9M annual GRS and improving style accuracy from 51.0% to 58.1%; led design and deployment of unified recommendation system raising pCTR AUC from 63% to 67% and pCVR AUC from 70% to 89%; optimized product rankings with profit aware exposure (PAE), generating \$8.1M annualized VCD gain and up to \$34M potential Browse impact; owned end-to-end system development from data strategy and model architecture to training and large-scale A/B testing
 - **Keystone AI** Bellevue, WA
Core AI Scientist Intern Jun 2025 - Jul 2025
 - **Tasks:** Developed a multi-echelon supply chain simulator, modeled consumer, store, and manufacturer interactions to forecast demand; analyzed cause and effect scenarios; contributed to the retrieval-based Seq2Seq architecture (MQCNN) for multi-horizon time series forecasting, integrated customer-level signals, leveraged positional encoding from transformer architectures in large language models (LLMs), contributed to a foundational supply chain forecasting model
 - **Amazon** Bellevue, WA
Research Scientist Intern: SCOT Topline Forecast May 2024 - Aug 2024
 - **Tasks:** Studied causal effects of speed on average selling price (ASP); used zip code level data to examine YoY correlations between different speed metrics and ASP in US AFN; applied the Bayesian VAR module to estimate effects on ASP rate and mix; implemented recursive assumption, sign restriction and heteroskedasticity on identification; estimated time-varying elasticities during major holidays (e.g., Christmas) and more price-sensitive around major sales

PUBLICATION / WORKING PAPER (GOOGLE SCHOLAR)

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1. **Yi Cui**, Yao Li, Jayson R. Miedema, Sharon N. Edmiston, Sherif W. Farag, James S. Marron, Nancy E. Thomas. *Region of Interest Detection in Melanocytic Skin Tumor Whole Slide Images - Nevus and Melanoma*. NeurIPS 2022 Workshop on Medical Imaging, Cancers 16(15), 2024. [Abstract][Paper][Codes]
 2. Andrii Babii*, **Yi Cui***, Thomas Walther*. *Macroeconomic Determinants of Realized Volatility - A Machine Learning Approach*, Working paper, 2025. (*equal contribution)
 3. Désiré Kédagni, Huan Wu, **Yi Cui**. *Robust Identification in Randomized Experiments with Noncompliance*, Working Paper, 2025. [Poster][Paper]

PATENTS / PROJECTS / HONORS

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- Yi Cui, National Patent S & F, First Inventor Health detector based on intelligent mobile terminal Feb 2019/Nov 2019
IPC Classification Number: A61B5/00 and A61B5/00, Patent Number: CN209611107U and CN109316169A
 - Project: Predicting the survival of patients, STOR 565: Advanced Machine Learning [Project] Jun 2021
 - Project: The Mathematical Contest in Modeling, MCM/ICM: Honorable Mention [Project] Jan 2018
 - Doctoral Dissertation; NABE Scholar; Summer Research Fellowship; Award of Excellent Student, First Prize Scholarship (top 1%); Second Prize in National Mathematical Modeling (CUMCM) (top 1%); Third Prize in Computer Programming Contest (top 5%); Silver Medal of National Mathematics Competition (top 1%); Morgan Stanley Investment Banking Early Insight Workshop Trainee, Goldman Sachs: GS Scholar Program Trainee 2016 - 2026

SKILLS

Technical Skills: Proficient in MATLAB, Python (Pytorch), R, C/C++, $\text{\LaTeX}$, and Julia

Fluent in English and Mandarin; CET-4: 667; CET-6: 600, TOEFL Writing: 30/30, IELTS: 7.0, GRE Math: 170/170